**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# PREDEFINED EXCEPTION

**PROGRAM:**

```
SQL>set serveroutput on;
declare
empid number(8);
emname varchar(10);
begin
empid:=&empid;
select empname into from employe where empno=empid;
dbms_output.put_line('The empoye name is:'||emname);
exception
when no_data_found then
dbms_output.put_line('data not found');
end;
/
```

**OUTPUT:**

```
Enter value for empid: 1
old   5: empid:=&empid;
new   5: empid:=1;
the employe name is:radha

PL/SQL procedure successfully completed.

SQL> /
Enter value for empid: 5
old   5: empid:=&empid;
new   5: empid:=5;
data not found

PL/SQL procedure successfully completed.
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

## USER DEFINED EXCEPTION

**PROGRAM:**

```
SQL> set serveroutput on;
 declare
 co number;
 exceeds_value exception;
 begin
 select count(*)into co from bankgrd;
 if co<5 then
insert into BANK values(&s_no,'&cust_name',&balance);
else
raise exceeds_value;
end if;
exception
when exceeds_value then
dbms_output.put_line('row will not be inserted');
end;
/
```

**OUTPUT:**

```
Enter value for s_no: 5
Enter value for cust_name: kaviya
Enter value for balance: 80000
old   7: insert into BANK values(&s_no,'&cust_name',&balance);
new   7: insert into BANK values(5,'kaviya',80000);

PL/SQL procedure successfully completed.
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# DATA DEFINITION LANGUAGE

## <u>CREATE TABLE COMMAND</u>

**Syntax:**

***SQL> Create table<table name>(fieldname1 datatype1(size),fieldname2 datatype2(size)….fieldname n datatype n(size));***

**Query:**

```
SQL> create table empxd(empno number(5), emp_name char(20), salary varchar(7), age number(4));

Table created.
```

**Created table:**

```
SQL> desc empxd;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                              NUMBER(5)
 EMP_NAME                                           CHAR(20)
 SALARY                                             VARCHAR2(7)
 AGE                                                NUMBER(4)
```

## <u>ALTER COMMAND</u>

The Alter command is associated with the following commands

- ADD
- MODIFY
- DROP

### USES OF ALTER COMMANDS

1. We can add a new field in an already created table
2. We can change the data type and size in a already created table
3. We can add integrity constraints to a field
4. We can drop integrity constraints of a field
5. We can drop a field (or) column from a table

### ADD FIELD

**Syntax:**

***SQL>alter table<table_name>add (fieldname1 datatype1(size)….fieldname n datatype n(size));***

**Query:**

```
SQL> alter table empxd add(job varchar(20));

Table altered.

SQL> desc empxd;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                              NUMBER(5)
 EMP_NAME                                           CHAR(20)
 SALARY                                             VARCHAR2(7)
 AGE                                                NUMBER(4)
 JOB                                                VARCHAR2(20)
```

## MODIFY FIELD

**Syntax:**

***SQL>altertable<table_name>modify (fieldname1 datatype1(size)……fieldname n datatype n(size));***

**Query:**

```
SQL> alter table empxd modify(emp_name char(30));

Table altered.

SQL> desc empxd;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                              NUMBER(5)
 EMP_NAME                                           CHAR(30)
 SALARY                                             VARCHAR2(7)
 AGE                                                NUMBER(4)
 JOB                                                VARCHAR2(20)
```

## DROP FIELD

**Syntax:**

***SQL> alter table<table_name>drop<column_name>;***

**Query:**

```
SQL> alter table empxd modify(emp_name char(30));

Table altered.

SQL> desc empxd;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                              NUMBER(5)
 EMP_NAME                                           CHAR(30)
 SALARY                                             VARCHAR2(7)
 AGE                                                NUMBER(4)
 JOB                                                VARCHAR2(20)
```

## TRUNCATE TABLE

**Syntax:**

*SQL> truncate table<table_name>;*

**Query:**

```
SQL> truncate table empxd;

Table truncated.

SQL> select * from empxd;

no rows selected
```

## RENAME TABLE

**Syntax:**

*SQL>rename<table_name> to <new_table_name>;*

**Query:**

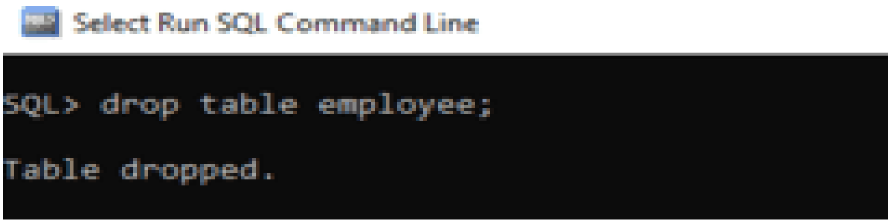
```
SQL> rename empxd to employeexd;

Table renamed.

SQL> desc employeexd;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                              NUMBER(5)
 EMP_NAME                                           CHAR(30)
 SALARY                                             VARCHAR2(7)
 AGE                                                NUMBER(4)
```

## DROP TABLE COMMAD

**Syntax:**

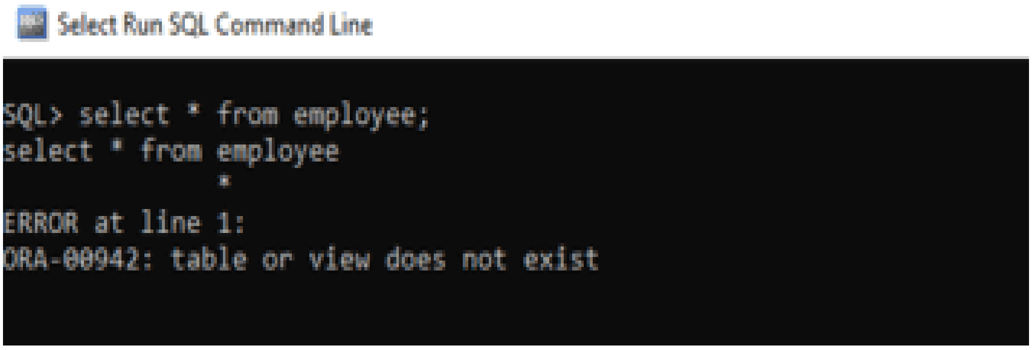
*SQL> drop table<table_name>;*

**Query:**


Select Run SQL Command Line

```
SQL> drop table employee;

Table dropped.
```

After drop the table will not be present,


Select Run SQL Command Line

```
SQL> select * from employee;
select * from employee
              *
ERROR at line 1:
ORA-00942: table or view does not exist
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# DATA MANIPULATION LANGUAGE

## INSERT

**Syntax:**

***SQL> insert into <table_name> values<value1,value2….value n>;***

**Query:**

```
SQL> insert into e1(e_id,e_name,e_sal)values(101,'raj',1000);

1 row created.

SQL> insert into e1(e_id,e_name,e_sal)values(102,'jessi',2000);

1 row created.

SQL> insert into e1(e_id,e_name,e_sal)values(103,'john',3000);

1 row created.
```

```
SQL> select * from e1;

      E_ID E_NAME          E_SAL
---------- ---------- ----------
       101 raj              1000
       102 jessi            2000
       103 john             3000
```

**UPDATE**

**Syntax:**

***SQL> update<table_name> set field_name=<expression>where [condition];***

**Query:**

```
SQL> update e1 set e_sal=5000 where e_name='john';

1 row updated.
```

**DELETE**

**Syntax:**

***SQL> delete from <table_name>where [condition];***

**Query:**

```
SQL> delete from e1 where(e_id=103);

1 row deleted.
```

```
SQL> select * from e1;

      E_ID E_NAME          E_SAL
---------- ---------- ----------
       101 raj              1000
       102 jessi            2000
```

REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A

# DATA QUERY LANGUAGE

## CREATE TABLE employe1:

```
SQL> create table employe1(empno number(5),ename char(10),job char(10),MGR number(5),sal number(5),deptno number(5),address char(10));


Table created.

SQL> insert into employe1(empno,ename,job,MGR,sal,deptno,address)values(7369,'guna','manager',7902,2000,20,'chennai');

1 row created.

SQL> insert into employe1(empno,ename,job,MGR,sal,deptno,address)values(7499,'kaushi','salesman',7617,1600,30,' ');

1 row created.
```

## TABLE STRUCTURE:

```
SQL> select*from employe1;

EMPNO                ENAME                JOB                         MGR
-------------------- -------------------- -------------------- ----------
       SAL     DEPTNO ADDRESS
---------- ---------- --------------------
7369                 guna                 manager                    7902
      2000         20 chennai

7499                 kaushi               salesman                   7617
      1600         30

7521                 ash                  clerk                      7839
      2975         10 trichy


EMPNO                ENAME                JOB                         MGR
-------------------- -------------------- -------------------- ----------
       SAL     DEPTNO ADDRESS
---------- ---------- --------------------
7342                 suman                salesman                   7893
      1200         20 hyderabad

76563                saran                manager                    7964
      4000         30
```

**SELECT**

**Syntax:**

*SQL>select <column_name>from<table_name>where[condition];*

**Query:**

```
SQL> select job from employe1 where empno=7369;

JOB
--------------------
manager

SQL>
```

**ORDER BY**

- **ASCENDING:**

**Syntax:**

*SQL>select<column_name>from<table_name>order by<column_name>;*

**Query:**

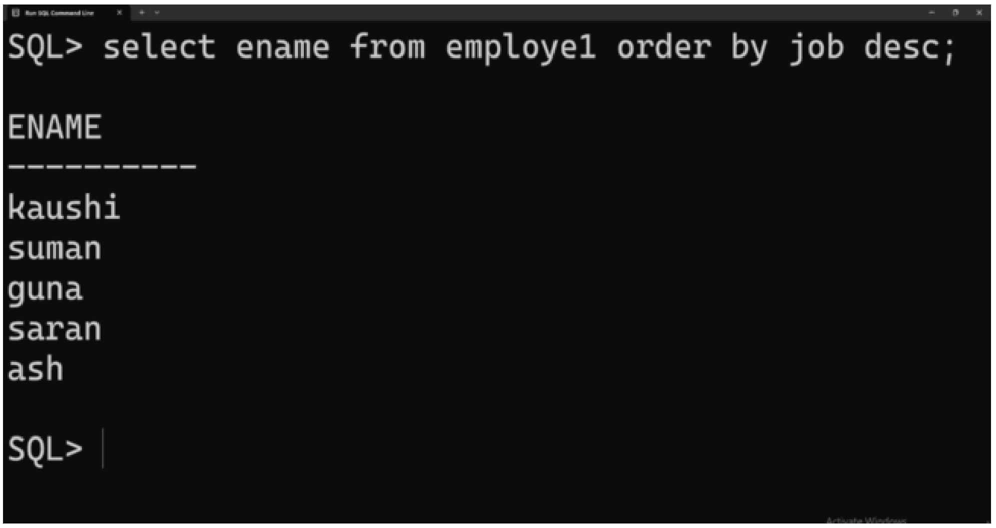
```
manager

SQL> select ename from employe1 order by job;

ENAME
----------
ash
saran
guna
kaushi
suman
```

- **DESCENDING:**

**Syntax:**

***SQL>select<column_name>from<table_name>order by<column_name>desc;***

**Query:**



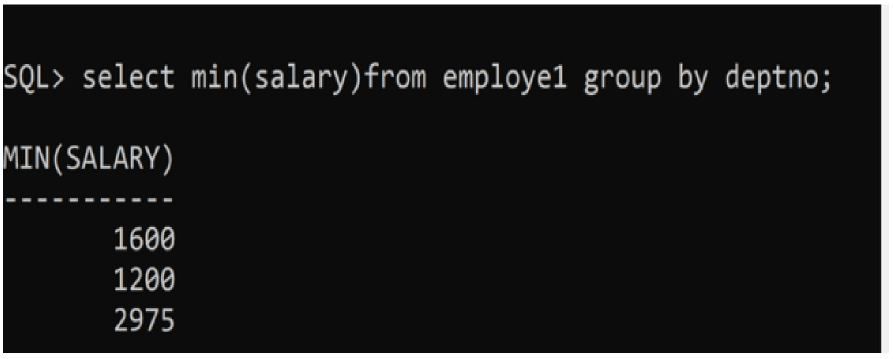
```
SQL> select ename from employe1 order by job desc;

ENAME
----------
kaushi
suman
guna
saran
ash

SQL>
```

**GROUP BY**

**Syntax:**

***SQL>select<column_name>from<table_name>order by<column_name>;***

**Query:**



```
SQL> select min(salary)from employe1 group by deptno;

MIN(SALARY)
-----------
       1600
       1200
       2975
```

## HAVING CLAUSE

**Syntax:**

***SQL>select<column_name>from<table_name>where[condition]group by column having[condition];***

**Query:**

```
SQL> select min(deptno)from employe1 group by salary having count(deptno)>=1;

MIN(DEPTNO)
-----------
         30
         20
         20
         10
         30

SQL>
```

## NULL

**Syntax:**

***SQL>select<column_name>from<table_name>where<column_name>is null;***

***SQL>select<column_name>from<table_name>where<column_name>is not null;***

**Query:**

```
SQL> select ename from employe1 where address=' ';

ENAME
--------------------
kaushi
saran
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# DATA CONTROL LANGUAGE

## CREATE TABLE employee1 AND employee2:

```
SQL> create table employee1(SLNO number(6),EMP_ID varchar(10),EMPNAME char(10),DESIGNATION char(12));

Table created.

SQL> insert into employee1(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(1,'19TD0301','jack','analyst');

1 row created.

SQL> insert into employee1(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(2,'19TD0302','harry','manager');

1 row created.
```

```
SQL> create table employee2(SLNO number(6),EMP_ID varchar(10),EMPNAME char(10),DESIGNATION char(12));

Table created.

SQL> insert into employee2(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(2,'19TD0302','harry','manager');

1 row created.

SQL> insert into employee2(SLNO,EMP_ID,EMPNAME,DESIGNATION)values(5,'19TD0305','morris','fresher');

1 row created.
```

## TABLE STRUCTURE:

```
mysql> select * from employee1;
+------+-----------+---------+--------------+
| SLNO | EMPID     | empname | designation  |
+------+-----------+---------+--------------+
|    1 | 22TD0055  | jack    | analyst      |
|    2 | 22TD0301  | harry   | manager      |
|    3 | 22TD0211  | john    | tester       |
|    4 | 22TD0050  | peter   | developer    |
+------+-----------+---------+--------------+
4 rows in set (0.00 sec)

mysql> select * from employee2;
+------+-----------+---------+--------------+
| SLNO | EMPID     | empname | designation  |
+------+-----------+---------+--------------+
|    2 | 22TD0302  | harry   | manager      |
|    5 | 22TD0305  | morris  | fresher      |
|    6 | 22TD0306  | ria     | junior       |
|    7 | 22TD0307  | henry   | teamleader   |
+------+-----------+---------+--------------+
4 rows in set (0.00 sec)
```

## COMMIT

**Syntax:**

*SQL> commit;*

**Query:**

```
      SLNO EMPID    EMPNAME  DESIGNATIO
---------- -------- -------- ----------
         1 19TD0301 jack     analyst
         2 19TD0302 harry    manager
         3 19TD0303 john     tester
         4 19TD0304 peter    developer

SQL> select*from employee2;

      SLNO EMPID    EMPNAME  DESIGNATIO
---------- -------- -------- ----------
         2 19TD0302 harry    manager
         5 19TD0305 morris   fresher
         6 19TD0306 ria      junior
         7 19TD0307 henry    teamleader

SQL> commit;

Commit complete.
```

## SAVEPOINT

**Syntax:**

*SQL> savepoint <name>;*

**Query:**

```
SQL> savepoint s;

Savepoint created.

SQL> select empname from employee1 union select empname from employee2;
```

**ROLLBACK**

**Syntax:**

*SQL> rollback;*

**Query:**

```
SQL> roll back;
Rollback complete.
SQL> select*from employee1;

      SLNO EMPID    EMPNAME  DESIGNATIO
---------- -------- -------- ----------
         1 19TD0301 jack     analyst
         2 19TD0302 harry    manager
         3 19TD0303 john     tester
         4 19TD0304 peter    developer

SQL> select*from employee2;

      SLNO EMPID    EMPNAME  DESIGNATIO
---------- -------- -------- ----------
         2 19TD0302 harry    manager
         5 19TD0305 morris   fresher
         6 19TD0306 ria      junior
         7 19TD0307 henry    teamleader
```

**GRANT**

**Syntax:**

***SQL>grant<select> on <table_name>to user;***

**Query:**

```
SQL> grant all on employee1 to system;

Grant succeeded.

SQL>
```

**UNION**

**Syntax:**

***SQL>select<column_name>from<table_name>union select <column_name>from<table_name>;***

**Query:**

```
SQL> select empname from employee1 union select empname from employee2;

EMPNAME
--------
harry
henry
jack
john
morris
peter
ria

7 rows selected.

SQL>
```

**UNIONALL**

**Syntax:**

***SQL> select<column_name>from<table_name>unionall select <column_name>from<table_name>;***

**Query:**

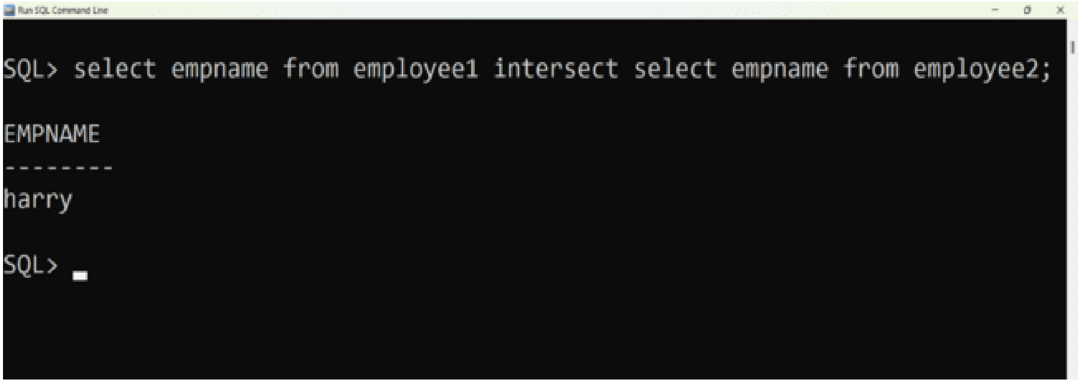
```
SQL> select empname from employee1 union all select empname from employee2;

EMPNAME
--------
jack
harry
john
peter
harry
morris
ria
henry

8 rows selected.

SQL>
```

**INTERSECT**

**Syntax:**

***SQL> select<column_name>from<table_name>intersect select <column_name>from<table_name>;***

**Query:**



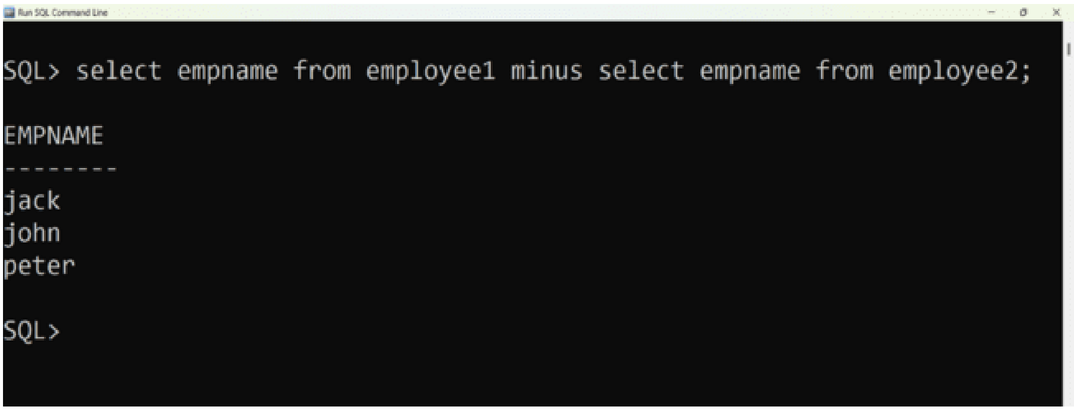
```
SQL> select empname from employee1 intersect select empname from employee2;

EMPNAME
--------
harry

SQL>
```

**MINUS**

**Syntax:**

***SQL> select<column_name>from<table_name>minus select <column_name>from<table_name>;***

**Query:**



```
SQL> select empname from employee1 minus select empname from employee2;

EMPNAME
--------
jack
john
peter

SQL>
```

# SET OPERATIONS

## CREATE TABLE employee AND department:

```
mysql> create table Employee(EMP_Name varchar(20), DEPT_NO int);
Query OK, 0 rows affected (0.01 sec)

mysql> insert into EMployee(EMP_Name, DEPT_NO) values('Guna', 10);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> create table Department(DEPT_Name varchar(20), DEPT_NO int);
Query OK, 0 rows affected (0.01 sec)

mysql> insert into Department(DEPT_Name, DEPT_NO) values('Manager', 10);
Query OK, 1 row affected (0.00 sec)
```

## TABLE STRUCTURE:

Table 1: Employee

```
SQL> select*from Employee;

EMP_NAME      DEPT_NO
---------- ----------
guna               10
suman              10
kaushi             20
vetri              30
ash                40
```

Table 2: Department

```
SQL> select*from department;

DEPT_NAME     DEPT_NO
---------- ----------
manager            10
salesman           20
staff              30
suman              10
guna               20

SQL>
```

**UNION:**

**Syntax:**

***SQL> select <column_name> from<table_name> union select <column_name>from<table_name>;***

**Query:**

```
SQL> select emp_name from employee union select dept_name fr
om department;

EMP_NAME
----------
ash
guna
kaushi
manager
salesman
staff
suman
vetri

8 rows selected.
```

**UNION ALL:**

**Syntax:**

***SQL> select <column_name> from<table_name> union all select<column_name>from<table_name>;***

**Query:**

```
SQL> select emp_name from employee union all select dept_name from depar
tment;

EMP_NAME
----------
guna
suman
kaushi
vetri
ash
manager
salesman
staff
suman
guna

10 rows selected.
```

**INTERSECTION:**

**Syntax:**

*SQL> select <column_name> from<table_name> intersect select<column_name>from<table_name>;*

**Query:**

```
SQL> select emp_name from employee int
ersect select dept_name from departmen
t;

EMP_NAME
----------
guna
suman
```

**MINUS:**

**Syntax:**

*SQL> select <column_name> from<table_name> minus select<column_name>from<table_name>;*

**Query:**

```
SQL> select emp_name from employee minus
select dept_name from department;

EMP_NAME
----------
ash
kaushi
vetri

SQL>
```

**CREATE TABLE employee_1:**

```
SQL> create table employee_1(EMP_NAME char(25),EMP_ID number(8),EMP_SALARY number(8),EMP_CITY char(15));

Table created.

SQL> insert into employee_1(EMP_NAME,EMP_ID,EMP_SALARY,EMP_CITY)values('mega',34857,40000,'pondy');

1 row created.
```

**TABLE STRUCTURE:**

Run SQL Command Line

```
SQL> select*from employee_1;

EMP_NAME   EMP_ID     EMP_SALARY EMP_CITY
---------- ---------- ---------- ----------
mega       34857      40000      pondy
ravi       34847      20000      chennai
mani       34237      25000      chennai
venai      34877      40000      madurai
rakul      34127      30000      pondy
raagu      34987      20000      madurai
priya      34097      30000      selam

7 rows selected.

SQL>
```

**BETWEEN:**

**Syntax:**

***SQL>select <column_name>from<table_name>where<column_name> BETWEEN value1 AND value2;***

**Query:**

**1)** List all the employee detail with salary in the range 20000 and 30000;

Run SQL Command Line

```
SQL> select emp_name,emp_id from employee_1 where emp_salary between 20000 and 30000;

EMP_NAME   EMP_ID
---------- ----------
ravi       34847
mani       34237
rakul      34127
raagu      34987
priya      34097

SQL> _
```

**NOT BETWEEN:**

**Syntax:**

***SQL>select<column_name>from<table_name>where<field_name>NOT BETWEEN value1 AND value2;***

**Query:**

**2)** List all the employee detail with salary not in the range 20000 and 30000;

Run SQL Command Line

```
SQL> select emp_name,emp_id from employee_1 where emp_salary not between 20000 and 30000;

EMP_NAME   EMP_ID
---------- ----------
mega       34857
venai      34877

SQL>
```

**IN:**

**Syntax:**

***SQL>select <column_name>from <table_name>WHERE<column_name>IN (value1,value2,…value n);***

**Query:**

**3)** List the employee detail whose city in pondy and chennai;

Run SQL Command Line

```
SQL> select emp_name,emp_id from employee_1 where emp_city in ('pondy','chennai');

EMP_NAME   EMP_ID
---------- ----------
mega       34857
ravi       34847
mani       34237
rakul      34127

SQL>
```

**NOT IN:**

**Syntax:**

*SQL>select <column_name>from<table_name>WHERE<field_name> NOT IN [condition];*

**Query:**

**4)** List the employee detail whose pondy not in chennai;

Run SQL Command Line

```
SQL> select emp_name,emp_id from employee_1 where emp_city not in ('pondy','chennai');

EMP_NAME   EMP_ID
---------- ----------
venai      34877
raagu      34987
priya      34097

SQL>
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# JOINS

1. **Inner Join**

- Equi Join
- Non-Equi Join
- Natural Join

2. **Outer Join**

- Left Outer Join
- Right Outer Join
- Full Outer Join

## CREATE TABLE stu AND dept:

```
SQL> create table stu(STU_ID varchar(6),S_NAME char(20),DEPT_ID number(5));

Table created.

SQL> insert into stu(STU_ID,S_NAME,DEPT_ID)valueinsert into stu(STU_ID,S_NAME,DEPT_ID)values(101,'raju',1);

1 row created.
```

```
SQL> create table dept(DEPT_ID varchar(6),D_NAME char(20));
Table created.

SQL> insert into dept(DEPT_ID,D_NAME)values(1,'cse');

1 row created.
```

## TABLE STRUCTURE:

## Table 1: stu

```
SQL> select * from stu;

S_ID   S_NAME   DEPT_ID
------ -------- ---------
101    raju     1
102    sai      2
103    pinky    1
104    hari     3
105    jessi    5
null   null     4

6 rows selected.
```

**Table 2: dept**

```
SQL> select * from dept;

DEPT_ID  D_NAME
-------- -------
1        cse
2        ece
3        eee
4        mech
5        null
```

**INNER JOIN**

## EQUI JOIN:

**Syntax:**

***SQL>select <column_name> from <table_name> where [condition];***

**Query:**

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu inner join dept on dept.dept_id=stu.dept_id;
```

```
S_ID   S_NAME   DEPT_ID  D_NAME
------ -------- -------- -------
101    raju     1        cse
102    sai      2        ece
103    pinky    1        cse
104    hari     3        eee
105    jessi    5        null
null   null     4        mech

6 rows selected.
```

**NON-EQUI JOIN:**

**Syntax:**

*SQL>select <column_name> from <table_name> where [condition];*

**Query:**

```
SQL> select s_name from stu where dept_id<4;
```

```
S_NAME
--------
raju
sai
pinky
hari
```

**SELF JOIN:**

**Syntax:**

*SQL>select <column_name> from <table_name> where [condition];*

**Query:**

```
SQL> select stu.s_name,dept.dept_id from stu,dept where dept.dept_id=stu.dept_id;
```

```
S_NAME   DEPT_ID
-------- --------
pinky    1
raju     1
sai      2
hari     3
null     4
jessi    5

6 rows selected.
```

**OUTER JOIN:**

**LEFT OUTER JOIN:**

**Syntax:**

***SQL>select <column_name> from <table 1> left outer join <table 2> on [condition];***

**Query:**

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu left outer join dept on dept.dept_id=stu.dept_id;
```

```
S_ID   S_NAME   DEPT_ID  D_NAME
------ -------- -------- -------
103    pinky    1        cse
101    raju     1        cse
102    sai      2        ece
104    hari     3        eee
null   null     4        mech
105    jessi    5        null

6 rows selected.
```

**RIGHT OUTER JOIN:**

**Syntax:**

***SQL>select <column_name> from <table 1> right outer join <table 2> on [condition];***

**Query:**

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu right outer join dept on dept.dept_id=stu.dept_id;
```

```
S_ID   S_NAME   DEPT_ID  D_NAME
------ -------- -------- -------
101    raju     1        cse
102    sai      2        ece
103    pinky    1        cse
104    hari     3        eee
105    jessi    5        null
null   null     4        mech

6 rows selected.
```

**FULL OUTER JOIN:**

**Syntax:**

***SQL>select <column_name> from <table 1> full outer join <table 2> on [condition]***

**Query:**

```
SQL> select stu.s_id,stu.s_name,dept.dept_id,dept.d_name from stu stu full outer join dept on dept.dept_id=stu.dept_id;
```

```
S_ID   S_NAME   DEPT_ID  D_NAME
------ -------- -------- -------
101    raju     1        cse
102    sai      2        ece
103    pinky    1        cse
104    hari     3        eee
105    jessi    5        null
null   null     4        mech

6 rows selected.
```

**CARTESIAN JOIN:**

**Syntax:**

***SQL>select table1.column1,table2.column2 from table1,table2;***

**Query:**

```
SQL> select * from stu,dept;
```

```
S_ID   S_NAME   DEPT_ID   DEPT_ID  D_NAME
------ -------- --------- -------- -------
101    raju     1         1        cse
101    raju     1         2        ece
101    raju     1         3        eee
101    raju     1         4        mech
101    raju     1         5        null
102    sai      2         1        cse
102    sai      2         2        ece
102    sai      2         3        eee
102    sai      2         4        mech
102    sai      2         5        null
103    pinky    1         1        cse

S_ID   S_NAME   DEPT_ID   DEPT_ID  D_NAME
------ -------- --------- -------- -------
103    pinky    1         2        ece
103    pinky    1         3        eee
103    pinky    1         4        mech
103    pinky    1         5        null
104    hari     3         1        cse
104    hari     3         2        ece
104    hari     3         3        eee
104    hari     3         4        mech
104    hari     3         5        null
105    jessi    5         1        cse
105    jessi    5         2        ece

S_ID   S_NAME   DEPT_ID   DEPT_ID  D_NAME
------ -------- --------- -------- -------
105    jessi    5         3        eee
105    jessi    5         4        mech
105    jessi    5         5        null
null   null     4         1        cse
null   null     4         2        ece
null   null     4         3        eee
null   null     4         4        mech
null   null     4         5        null

30 rows selected.
```

REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A

# VIEWS

## CREATE TABLE student:

```
SQL> create table student(STUD_ID number(5),STUD_NAME char(25),CGPA float(5),DEPT_ID varchar(5));

Table created.

SQL> insert into student values(100,'Saro',8.1,1);

1 row created.

SQL> insert into student values(101,'Jayaprada',8.5,1);

1 row created.
```

## TABLE STRUCTURE:

```
mysql> select * from students;
+---------+-------------+------+---------+
| STUD_ID | STUD_NAME   | CGPA | DEPT_ID |
+---------+-------------+------+---------+
|     100 | Saro        |  8.1 |       1 |
|     101 | Jayaprakash |  8.5 |       1 |
|     102 | Priya       |  8.6 |       1 |
|     103 | Vicky       |  7.6 |       1 |
|     104 | Ram         |  7.5 |       1 |
|     105 | Mandy       |  8.4 |       1 |
+---------+-------------+------+---------+
6 rows in set (0.00 sec)
```

## CREATING VIEWS

**Syntax:**

***SQL>create view <view name> as select column1, column2...column n from <table name> where [condition];***

**Query:**

View is created for the table,

```
mysql> CREATE VIEW students_dept1 AS
    -> SELECT STUD_ID, STUD_NAME, CGPA, DEPT_ID
    -> FROM students
    -> WHERE DEPT_ID = 1;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM students_dept1;
+---------+-------------+------+---------+
| STUD_ID | STUD_NAME   | CGPA | DEPT_ID |
+---------+-------------+------+---------+
|     100 | Saro        |  8.1 |       1 |
|     101 | Jayaprakash |  8.5 |       1 |
|     102 | Priya       |  8.6 |       1 |
|     103 | Vicky       |  7.6 |       1 |
|     104 | Ram         |  7.5 |       1 |
|     105 | Mandy       |  8.4 |       1 |
+---------+-------------+------+---------+
6 rows in set (0.00 sec)
```

**INSERT:**

**Syntax:**

***SQL>insert into <view_name>(fieldname1,fieldname2….filedname n)values(value1,value2….value n);***

**Query:**

```
mysql> select * from CSE;
+---------+-------------+------+---------+
| stud_id | stud_name   | cgpa | dept_id |
+---------+-------------+------+---------+
|     100 | Saro        |  8.1 |       1 |
|     101 | JayaPrakash |  8.5 |       1 |
|     102 | Mandy       |  7.5 |       1 |
|     103 | Anisha      |    9 |       1 |
+---------+-------------+------+---------+
4 rows in set (0.00 sec)
```

Table after inserting in view,

```
mysql> select * from students;
+---------+-------------+------+---------+
| STUD_ID | STUD_NAME   | CGPA | DEPT_ID |
+---------+-------------+------+---------+
|     100 | Saro        |  8.1 |       1 |
|     101 | Jayaprakash |  8.5 |       1 |
|     102 | Priya       |  8.6 |       1 |
|     103 | Vicky       |  7.6 |       1 |
|     104 | Ram         |  7.5 |       1 |
|     105 | Mandy       |  8.4 |       1 |
|     106 | Anmol       |    9 |       2 |
|     107 | Akash       |  7.5 |       2 |
+---------+-------------+------+---------+
8 rows in set (0.00 sec)
```

**UPDATING A VIEW:**

**Syntax:**

*SQL>update <view_name> where [condition];*

**Query:**

```
mysql> update CSE set cgpa=9 where stud_id = 100;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from CSE;
+---------+-------------+------+---------+
| stud_id | stud_name   | cgpa | dept_id |
+---------+-------------+------+---------+
|     100 | Saro        |    9 |       1 |
|     101 | JayaPrakash |  8.5 |       1 |
|     102 | Mandy       |  7.5 |       1 |
|     103 | Anisha      |    9 |       1 |
+---------+-------------+------+---------+
4 rows in set (0.00 sec)
```

Table after updating the view,

```
mysql> select * from students;
+---------+-------------+------+---------+
| STUD_ID | STUD_NAME   | CGPA | DEPT_ID |
+---------+-------------+------+---------+
|     100 | Saro        |    9 |       1 |
|     101 | Jayaprakash |  8.5 |       1 |
|     102 | Mandy       |  7.5 |       1 |
|     103 | Anisha      |    9 |       1 |
|     104 | Ram         |  7.5 |       2 |
|     105 | Mandy       |  8.4 |       2 |
|     106 | Anmol       |    9 |       2 |
|     107 | Akash       |  7.5 |       2 |
+---------+-------------+------+---------+
8 rows in set (0.00 sec)
```

**DELETE:**

**Syntax:**

*SQL> delete from <view_name> where [condition];*

**Query:**

```
mysql> delete from CSE where stud_id = 103;
Query OK, 1 row affected (0.00 sec)

mysql> select * from CSE;
+---------+-------------+------+---------+
| stud_id | stud_name   | cgpa | dept_id |
+---------+-------------+------+---------+
|     100 | Saro        |    9 |       1 |
|     101 | JayaPrakash |  8.5 |       1 |
|     102 | Mandy       |  7.5 |       1 |
+---------+-------------+------+---------+
3 rows in set (0.00 sec)
```

Table after deleting in view,

```
mysql> select * from students;
+---------+-------------+------+---------+
| STUD_ID | STUD_NAME   | CGPA | DEPT_ID |
+---------+-------------+------+---------+
|     100 | Saro        |    9 |       1 |
|     101 | Jayaprakash |  8.5 |       1 |
|     102 | Mandy       |  7.5 |       1 |
|     104 | Ram         |  7.5 |       2 |
|     105 | Mandy       |  8.4 |       2 |
|     106 | Anmol       |    9 |       2 |
|     107 | Akash       |  7.5 |       2 |
+---------+-------------+------+---------+
7 rows in set (0.00 sec)
```

**DROP:**

**Syntax:**

*SQL>drop view<view_name>;*

**Query**:

SQL Plus

```
SQL> drop view CSE;

View dropped.
```

After the view is dropped the table will be remind the same,

```
mysql> select * from CSE;
ERROR 1146 (42S02): Table 'jazxd.cse' doesn't exist
mysql> select * from students;
+---------+-------------+------+---------+
| STUD_ID | STUD_NAME   | CGPA | DEPT_ID |
+---------+-------------+------+---------+
|     100 | Saro        |    9 |       1 |
|     101 | Jayaprakash |  8.5 |       1 |
|     102 | Mandy       |  7.5 |       1 |
|     104 | Ram         |  7.5 |       2 |
|     105 | Mandy       |  8.4 |       2 |
|     106 | Anmol       |    9 |       2 |
|     107 | Akash       |  7.5 |       2 |
+---------+-------------+------+---------+
7 rows in set (0.00 sec)
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# SUB QUERIES

## CREATE TABLE student AND database:

```
mysql> create table data(STU_NAME varchar(10), STU_ROLL int, STU_LOCATE varchar(10));
Query OK, 0 rows affected (0.01 sec)

mysql> insert into data(STU_NAME, STU_ROLL, STU_LOCATE) values('Ravi', 1123, 'Chennai');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> create table STUdata(STU_NAME varchar(10), STU_ROLL int, STU_SEC varchar(10));
Query OK, 0 rows affected (0.01 sec)

mysql> insert into STUdata(STU_NAME, STU_ROLL, STU_SEC) values('Ravi', 1123, 'A');
Query OK, 1 row affected (0.00 sec)
```

## TABLE STRUCTURE:

Run SQL Command Line

```
SQL> select*from database;

STU_NAME   STU_ROLL   STU_LOCATI
---------- ---------- ----------
RAVI       1123       chennai
RAJ        1124       madurai
RAM        1125       salem
SAM        1126       theni
MANI       1127       chennai

SQL> select*from student;

STU_NAME   STU_ROLL   STU_SEC
---------- ---------- ----------
RAVI       1123       A
RAJ        1124       B
RAM        1125       A

SQL>
```

**SELECT:**

**Syntax:**

***SQL> SELECT column_name***

***FROM table_name***

***WHERE column_name expression operator***

***( SELECT column_name from table_name WHERE ... );***

**Query:**

Run SQL Command Line

```
SQL> select stu_name,stu_location from database where stu_roll in (select stu_roll from student where stu_sec='A');

STU_NAME   STU_LOCATI
---------- ----------
RAVI       chennai
RAM        salem

SQL>
```

**INSERT**

**Syntax:**

*SQL> INSERT INTO table_name (column1, column2, column3....)*

*SELECT **

*FROM table_name*

*WHERE VALUE OPERATOR*

**Query:**

Run SQL Command Line

```
SQL> select*from student1;

NAME       ROLL_NO        LOCATION       PHONENUMBER
---------- -------------- -------------- ------------
RAVI       1212           CHENNAI        9003733628
MOTHI      1232           THENI          7397080037
MADHAN     1242           MADURAI        9790585064
MORISH     1282           THENI          1234080037

SQL> select*from student2;

NAME       ROLL_NO        LOCATION       PHONENUMBER
---------- -------------- -------------- ------------
RAHUL      4212           KERALA         9093433628
MASHA      1242           MADURAI        9792285064
MONISHA    1442           CHENNAI        9792285234

SQL> insert into student1 select*from student2;

3 rows created.

SQL> select *from student1;

NAME       ROLL_NO        LOCATION       PHONENUMBER
---------- -------------- -------------- ------------
RAVI       1212           CHENNAI        9003733628
MOTHI      1232           THENI          7397080037
MADHAN     1242           MADURAI        9790585064
MORISH     1282           THENI          1234080037
RAHUL      4212           KERALA         9093433628
MASHA      1242           MADURAI        9792285064
MONISHA    1442           CHENNAI        9792285234

7 rows selected.

SQL>
```

**DELETE:**

**Syntax:**

***SQL>DELETE FROM TABLE_NAME***

***WHERE VALUE OPERATOR***

***(SELECT COLUMN_NAME***

***FROM TABLE_NAME***

***WHERE condition);***

```
SQL> delete from student2 where roll_no in (select roll_no from student1 where location='CHENNAI');

1 row deleted.

SQL> select*from student2;

NAME       ROLL_NO      LOCATION     PHONENUMBER
---------- ------------ ------------ -----------
MASHA      1242         MADURAI      9792285064
RAKUL      1542         KERALA       1234564552

SQL>
```

**UPDATE**

**Syntax:**

***SQL> UPDATE table***

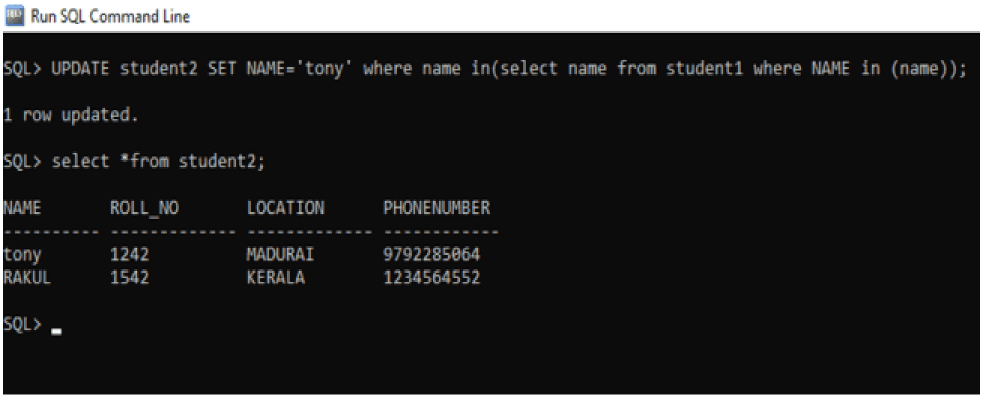
***SET column_name = new_value***

***WHERE VALUE OPERATOR***

***(SELECT COLUMN_NAME***

***FROM TABLE_NAME***

***WHERE condition);***



```
Run SQL Command Line

SQL> UPDATE student2 SET NAME='tony' where name in(select name from student1 where NAME in (name));

1 row updated.

SQL> select *from student2;

NAME       ROLL_NO       LOCATION      PHONENUMBER
---------- ------------- ------------- ------------
tony       1242          MADURAI       9792285064
RAKUL      1542          KERALA        1234564552

SQL>
```

# ADDITION OF TWO NUMBERS

**PROGRAM:**

**SQL>**set serveroutput on;

```
declare
x integer;
y integer;
z integer;
begin
x:=&x;
y:=&y;
z:=x+y;
dbms_output.put_line('The sum of two numbers is'||z);
end;
/
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# FACTORIAL OF A NUMBER

**PROGRAM:**

**SQL>** set serveroutput on;

```
declare
f number:=1;
i number;
n number:=&n;
begin
for i in 1..n
loop
f:=f*i;
end loop;
dbms_output.put_line(‘the factorial of given number is:’||f);
end;
/
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# SUM OF SERIES

**PROGRAM:**

**SQL>** set serveroutput on;

```
declare
n number:=&n;
i number;
begin
i:=n*(n+1);
n:=i/2;
dbms_output.put_line('the sum of series of given number is:'||n);
end;
/
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# FIBONACCI SERIES

**PROGRAM:**

**SQL>**set serveroutput on;

```
declare
i number;
c number;
n number:=&n;
a number:=-1;
b number:=1;
begin
for i in 1..n
loop
c:=a+b;
dbms_output.put_line(c);
a:=b;
b:=c;
end loop;
end;
/
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# PALINDROME SERIES

**PROGRAM:**

```
SQL>set serveroutput on;
    declare
    len number;
    a integer;
    str1 varchar(10):='&str1';
    str2 varchar(10);
    begin
    len:=length(str1);
    a:=len;
    for i in 1..a
    loop
    str2:=str2||substr(str1,len,1);
    len:=len-1;
    end loop;
    if(str1=str2)then
    dbms_output.put_line(str1||'   is a palindrome');
    else
    dbms_output.put_line(str1||' is not a palindrome');
    end if;
    end;
    /
```

**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# PACKAGES

**SOURCE CODE:**

**PACKAGE EXCEPTION:**

```
create or replace package packs is
procedure display;
function prime(n number) return varchar;
function odd(n number) return varchar;
function positive(n number)return varchar;
end packs;
```

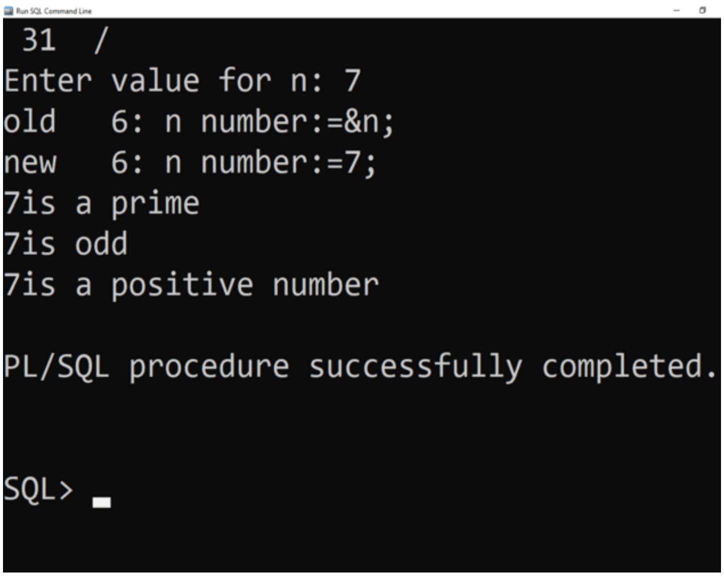
**PACKAGE BODY:**

```
create or replace package body packs as
procedure display is
avarchar(15);
m number;
begin
m:=&m;
a:=prime(m);
if a=prime(m)then
dbms_output.put_line(m||'is a prime');
else
dbms_output.put_line(m||'is not prime');
end if;
```

```
a:=odd(m);
if a='odd' then
dbms_output.put_line(m||'is odd');
else
dbms_output.put_line(m||'is even');
end if;
a:=positive(m);
if a='positive' then
dbms_output.put_line(m||'is a positive number');
else
dbms_output.put_line(m||'is a negative number');
end if;
end display;
function prime(n number) return varchar is
i number;
flag number;
begin
flag:=0;
for i in 2..n/2
loop
if mod(n,i)=0 then
flag:=1
return 'not prime';
end if;
end loop;
if flag=0 then
return 'prime';
end if;
```

```
end prime;
function odd(n number) return varchar is
begin
if n>0 then
return 'positive';
else
return 'negative';
end if;
end positive;
end packs;
```

**OUTPUT:**



**REG NO: 22TD0075 NAME: SANDEEP G YEAR/SEM/SEC: Y3/S5/A**

# CURSOR

**SOURCE CODE:**

**DELETING THE RECORD USING CURSOR**

```
SQL> declare
        cursor c is select * from vechicle1;
        a c % rowtype;
        n number:=0;
        begin
        open c;
        loop
        fetch c into a;
        exit when c% notfound;
        ifa.vehicle_no=521 then
        delete from vehicle1 where vehicle_no=521;
        n:=n+1;
        end if;
        end loop;
        dbms_output.put_line('deleted record'||n);
        dbms_output.put_line('remaining records'||(c % rowcount-n));
        close c;
        commit;
        end;
          /
```

**OUTPUT:**

```
VEHICLE_NO VEHIC SOURCE     DESTINATIO  SEAT_COST   ARR_TIME
---------- ----- ---------- ---------- ---------- ----------
       521 duke  pondy      cuddalore          52         12
       623 prtc  pondy      banglore           35          2
       632 mnc   bangalore  chennai            23          1

SQL>
```

```
PL/SQL procedure successfully completed.

SQL> select * from vechicle1;

VEHICLE_NO VEHICLE SOURCE     DESTINATIO  SEAT_COST   ARR_TIME
---------- ------- ---------- ---------- ---------- ----------
       623 prtc    pondy      banglore           35          2
       632 mnc     bangalore  chennai            23          1

SQL>
```